



THE UNIVERSITY OF DA NANG - UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

FINAL TERM REPORT
EMBEDDED SYSTEM

PROJECT: SMART LIGHT

Lecturer: Ho Viet Viet

Group: HAP

- 1. Nguyen Van An**
- 2. Le Huy Hoang**
- 3. Pham Thi Phuong**

Da Nang, 2025

CONTENT:

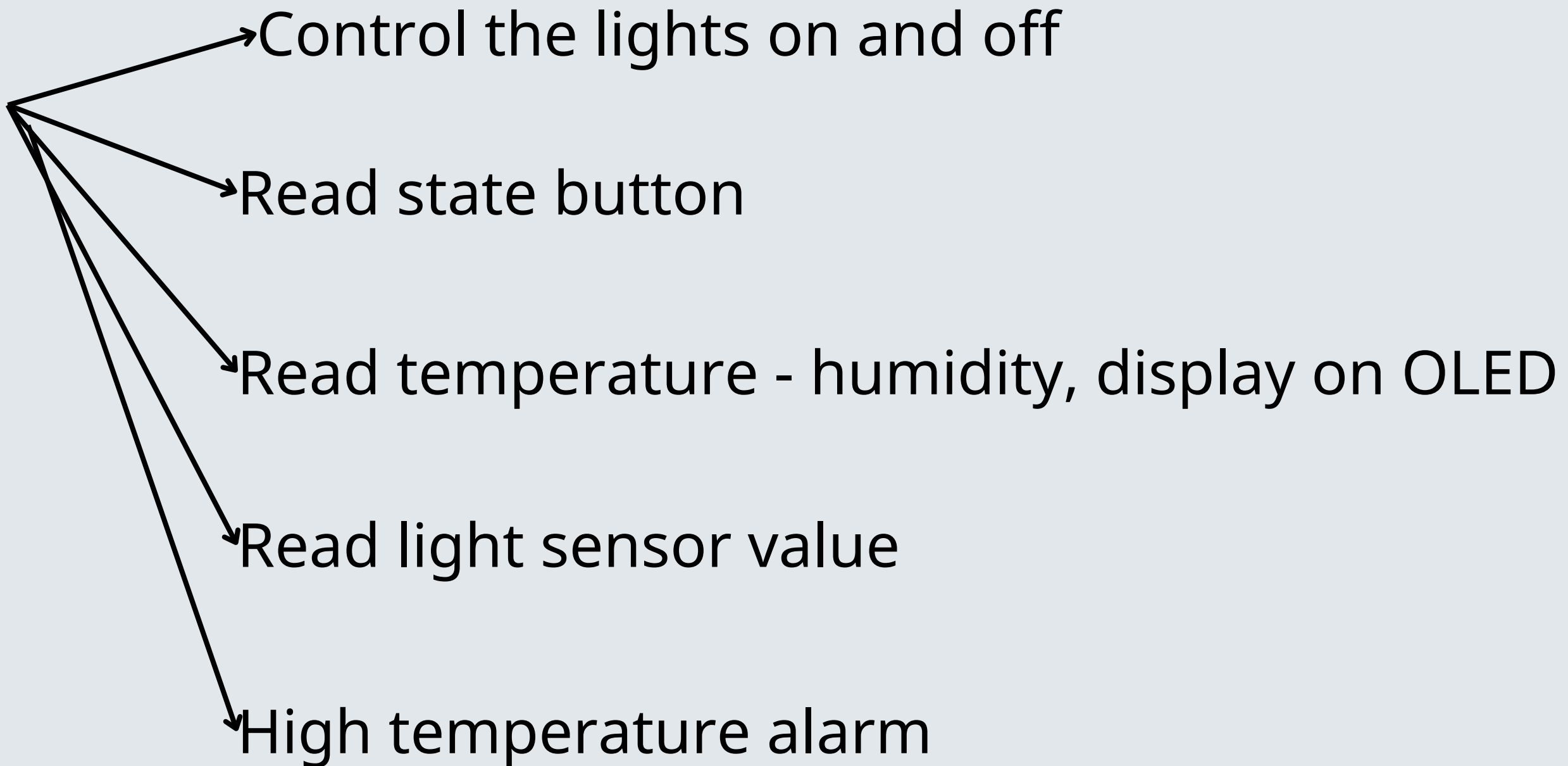
- 1. Introduce**
- 2. Systems analysis**
- 3. Design & Implementation**
- 4. Evaluate the results**

1. Introduce

1.1 Introduce

- Build smart light system based on ESP32-C3 + FreeRTOS

- Function



```
graph LR; A(( )) --> B[Control the lights on and off]; A --> C[Read state button]; A --> D[Read temperature - humidity, display on OLED]; A --> E[Read light sensor value]; A --> F[High temperature alarm];
```

Control the lights on and off

Read state button

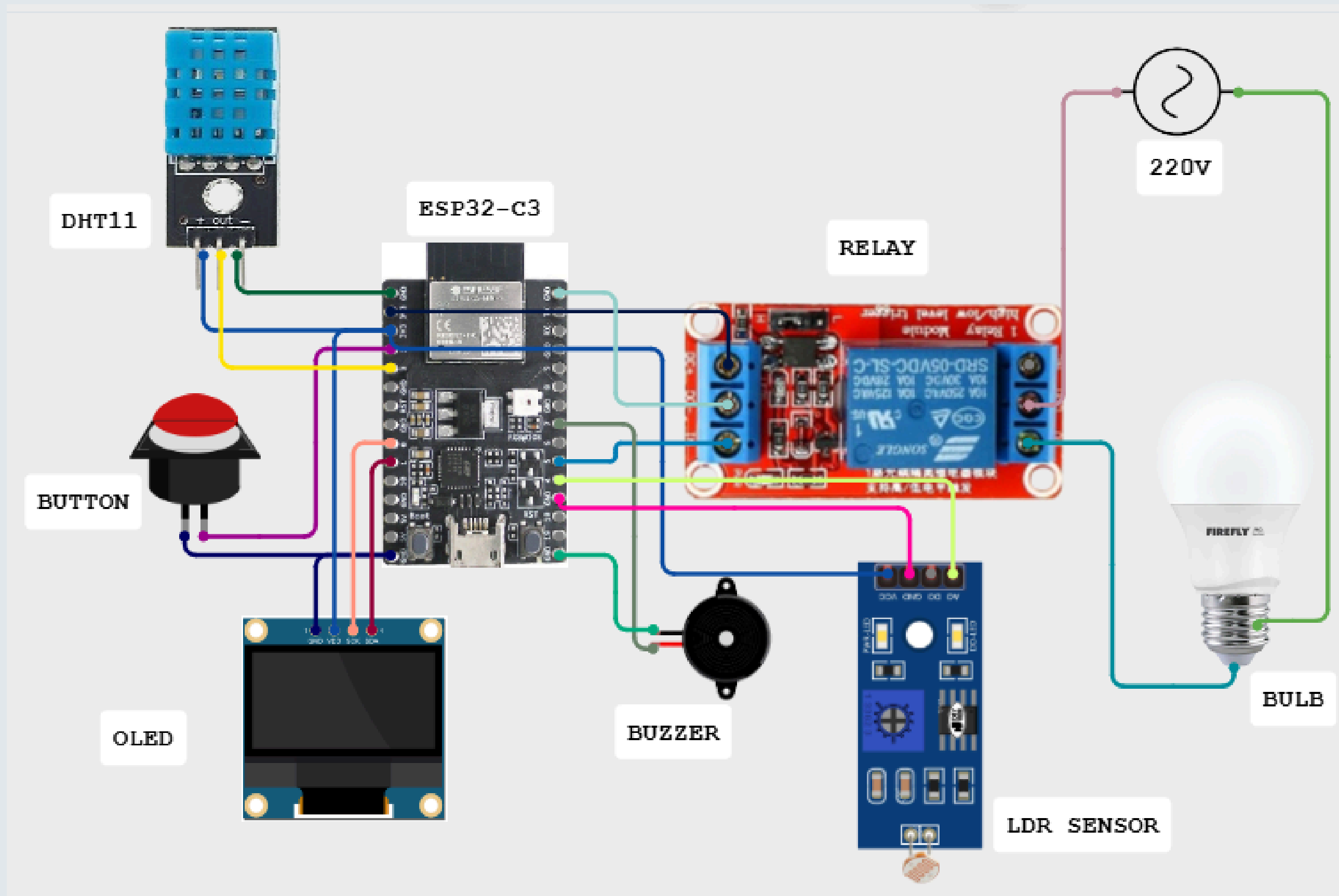
Read temperature - humidity, display on OLED

Read light sensor value

High temperature alarm

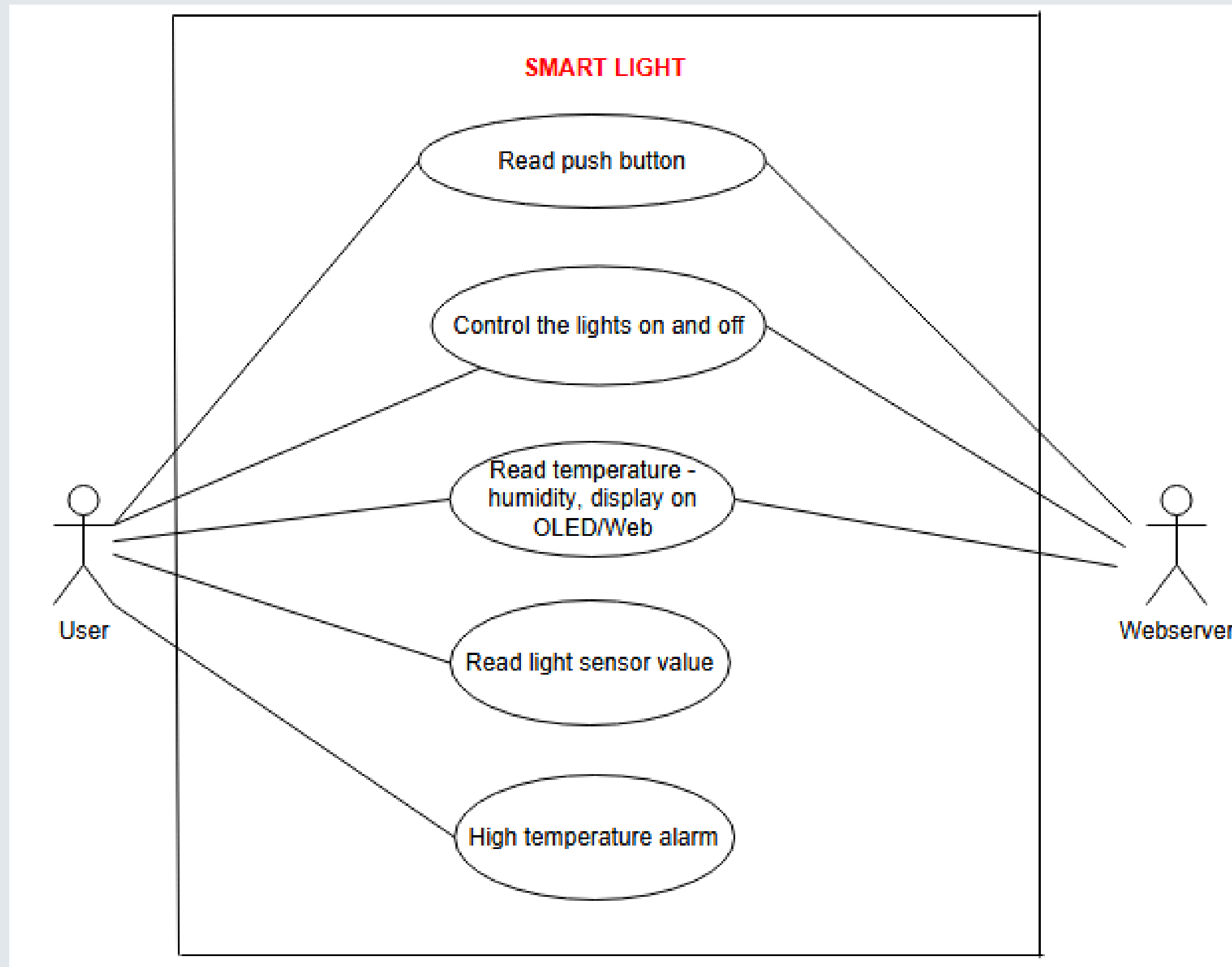
1. Introduce

1.2 Connection diagram



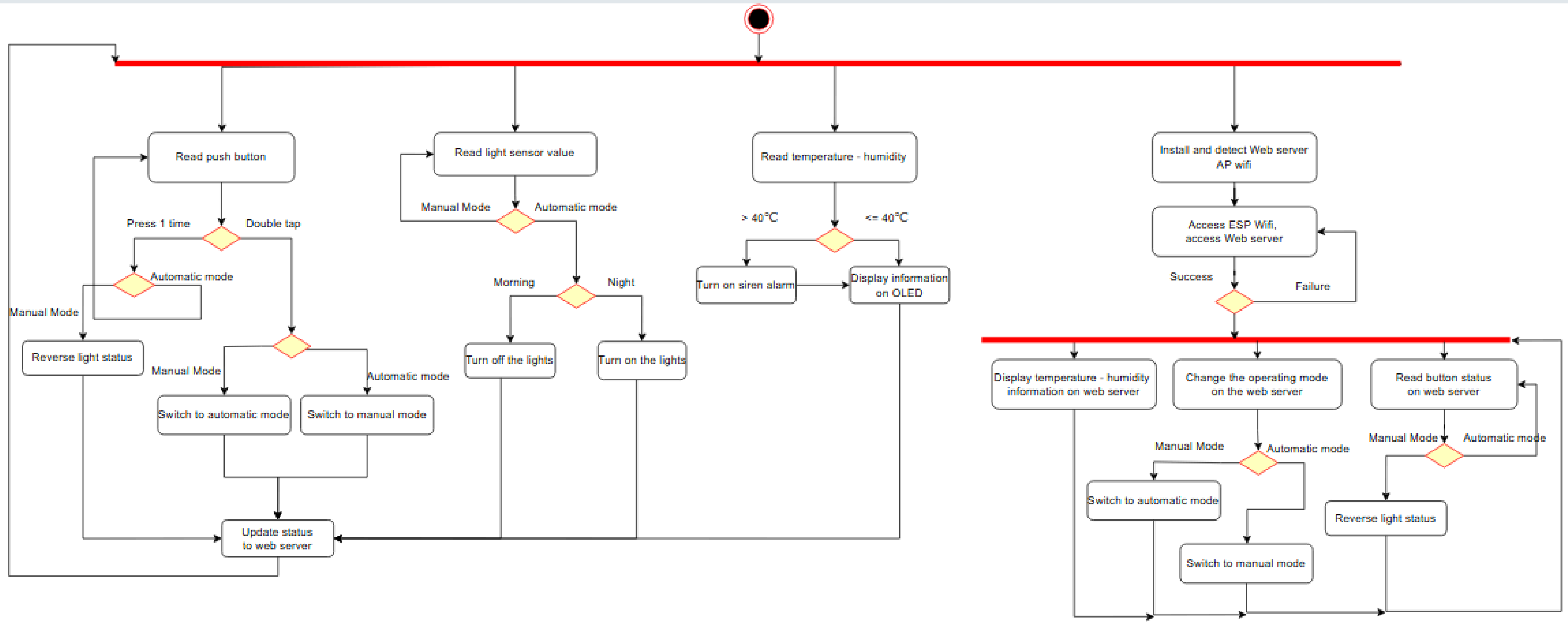
2. Systems analysis

2.1 Use case diagram



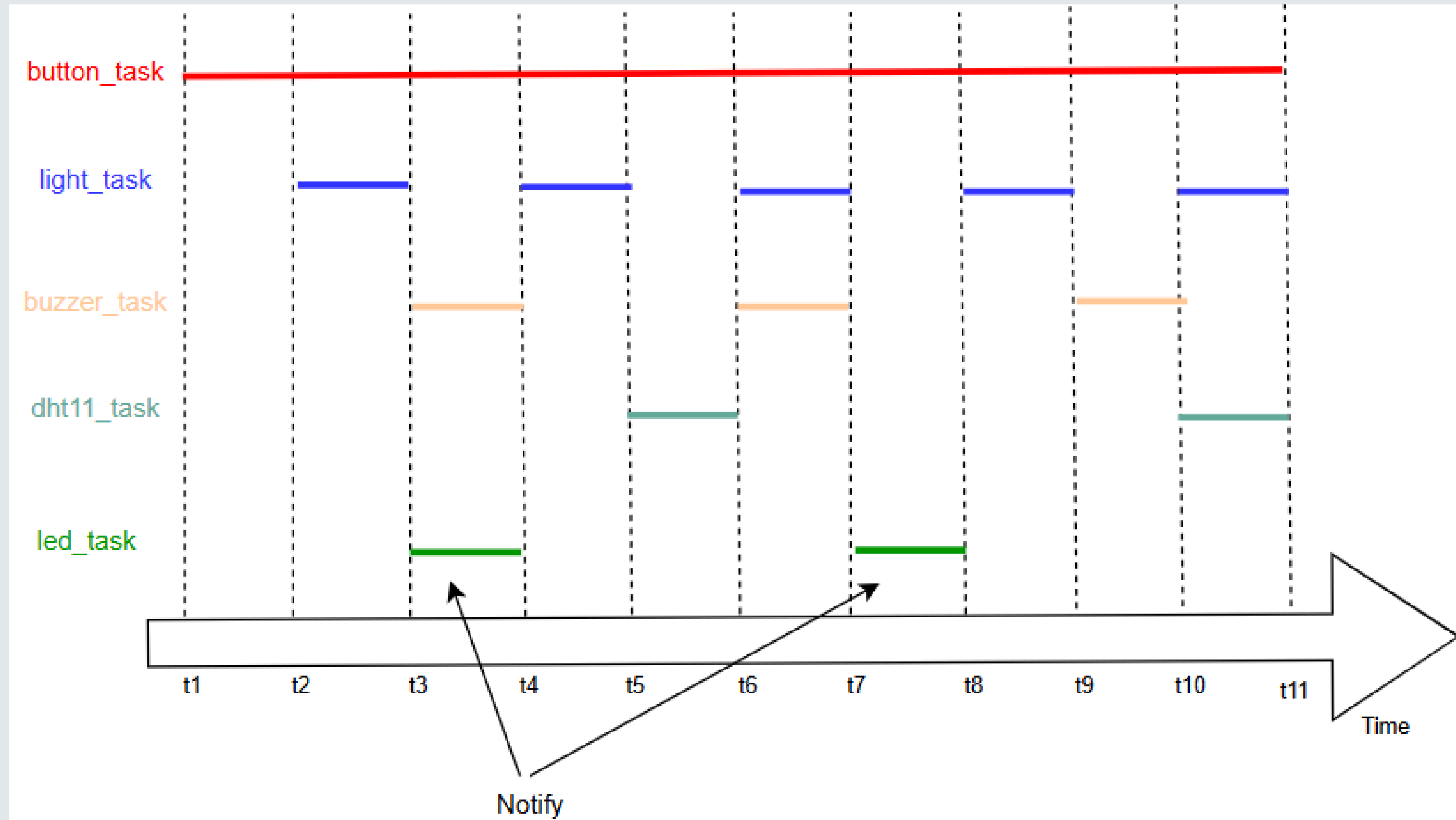
2. Systems analysis

2.2 Activity diagram



2. Systems analysis

2.3 Timing diagram



3. Design & Implementation

3.1 Task

- dht11_task: Read DHT11, update data, update OLED and report overheat.
- button_task: Handle single/double click to change mode or turn on/off light.
- light_task: Read LDR and decide to turn on/off light.
- led_task: Receive notification and control LED + update status.
- buzzer_task: Check for overheat event and turn on/off buzzer.

```
xTaskCreate(led_task, "led_task", 2048, NULL, 4, &ledTaskHandle);  
xTaskCreate(dht11_task, "dht11_task", 4096, NULL, 5, NULL);  
xTaskCreate(button_task, "button_task", 2048, NULL, 6, NULL);  
xTaskCreate(light_task, "light_task", 2048, NULL, 5, NULL);  
xTaskCreate(buzzer_task, "buzzer_task", 2048, NULL, 5, NULL);
```


3. Design & Implementation

3.2 Mutex

- Mutex is used to protect shared variables (mode, led_state_global, DHT11 data) from conflicts when multiple tasks access them at the same time.

```
static SemaphoreHandle_t data_mutex = NULL;
```

```
data_mutex = xSemaphoreCreateMutex();  
if (data_mutex == NULL) {  
    printf("Failed to create data_mutex\n");  
    while (1) vTaskDelay(1000 / portTICK_PERIOD_MS);  
}  
xSemaphoreTake(data_mutex, portMAX_DELAY);  
dht11_sensor.temperature = curr_temp;  
dht11_sensor.humidity = curr_hum;  
xSemaphoreGive(data_mutex);
```

```
if (m == 0) {    // Manual  
  
    xSemaphoreTake(data_mutex, portMAX_DELAY);  
    int new_state = !led_state_global;  
    xSemaphoreGive(data_mutex);
```

3. Design & Implementation

3.3 Event Groups

- Event Group is used to mark and monitor system states (Auto mode, LED ON, Over Temperature) so that other tasks can execute based on it.

```
sys_event_group = xEventGroupCreate();
if (sys_event_group == NULL) {
    printf("Failed to create sys_event_group\n");
    while (1) vTaskDelay(1000 / portTICK_PERIOD_MS);
}
```

```
void buzzer_task(void *pvParameter) {
    while (1) {
        EventBits_t bits = xEventGroupGetBits(sys_event_group);
        if (bits & BIT_OVER_TEMP) {
            gpio_set_level(BUZZER_GPIO, 1); // on
        } else {
            gpio_set_level(BUZZER_GPIO, 0); // off
        }
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}
```

```
        gpio_set_level(LED_GPIO, led_state_global);

        if (led_state_global)
            xEventGroupSetBits(sys_event_group, BIT_LED_ON);
        else
            xEventGroupClearBits(sys_event_group, BIT_LED_ON);
    }
}
```

3. Design & Implementation

3.4 Task Notifications

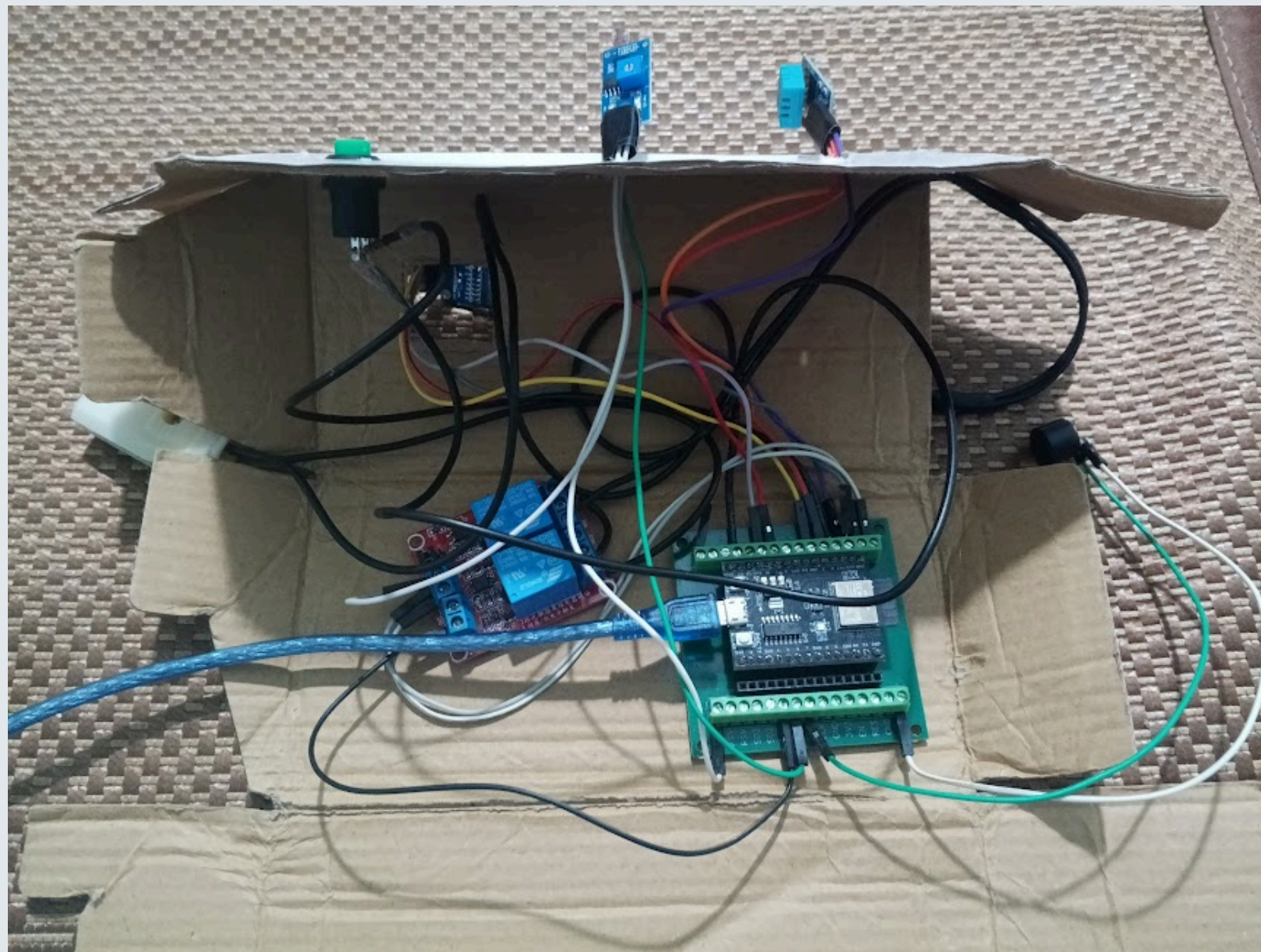
- Task Notification is used to send LED on/off commands quickly and directly from other tasks to led_task without queuing.

```
    if (ledTaskHandle != NULL) {  
        // notify with new_state (0 hoặc 1)  
        xTaskNotify(ledTaskHandle, (uint32_t)new_state, eSetValueWithOverwrite);  
    }  
}  
  
    if (ledTaskHandle != NULL) {  
        xTaskNotify(ledTaskHandle, (uint32_t)new_state, eSetValueWithOverwrite);  
    }  
}  
}  
  
    if (ledTaskHandle != NULL) {  
        xTaskNotify(ledTaskHandle, (uint32_t)led_state, eSetValueWithOverwrite);  
    }  
}  
  
if (xTaskNotifyWait(0x00, 0xFFFFFFFF, &led_state, portMAX_DELAY) == pdTRUE) {  
  
    xSemaphoreTake(data_mutex, portMAX_DELAY);  
    led_state_global = (int)led_state;  
    xSemaphoreGive(data_mutex);  
}
```


4. Evaluate the results

4.1 Result

- Stable operating system: Manual, Auto (LDR), Web UI, OLED, Buzzer all meet design requirements

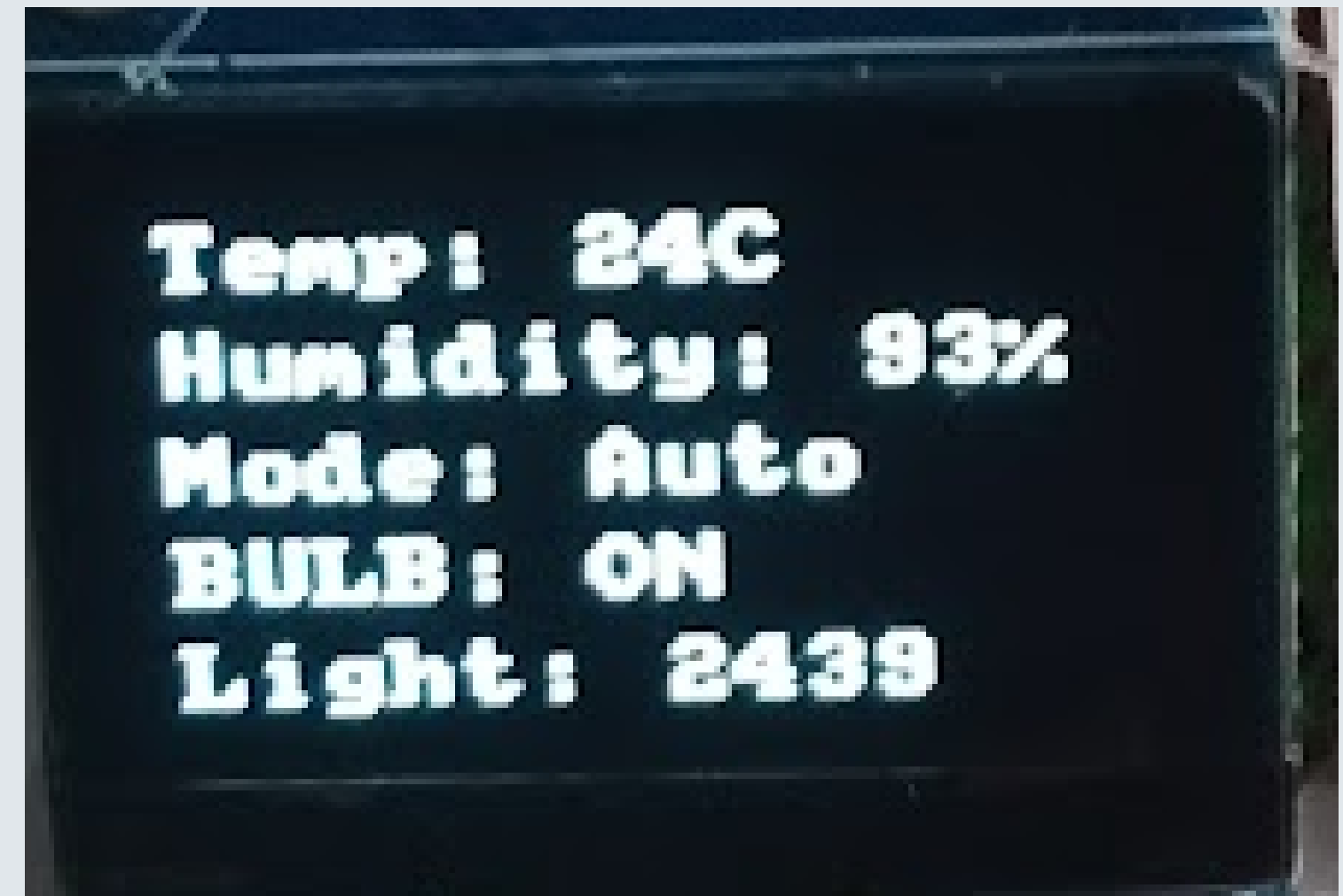


4. Evaluate the results

4.1 Result

OLED Display Information:

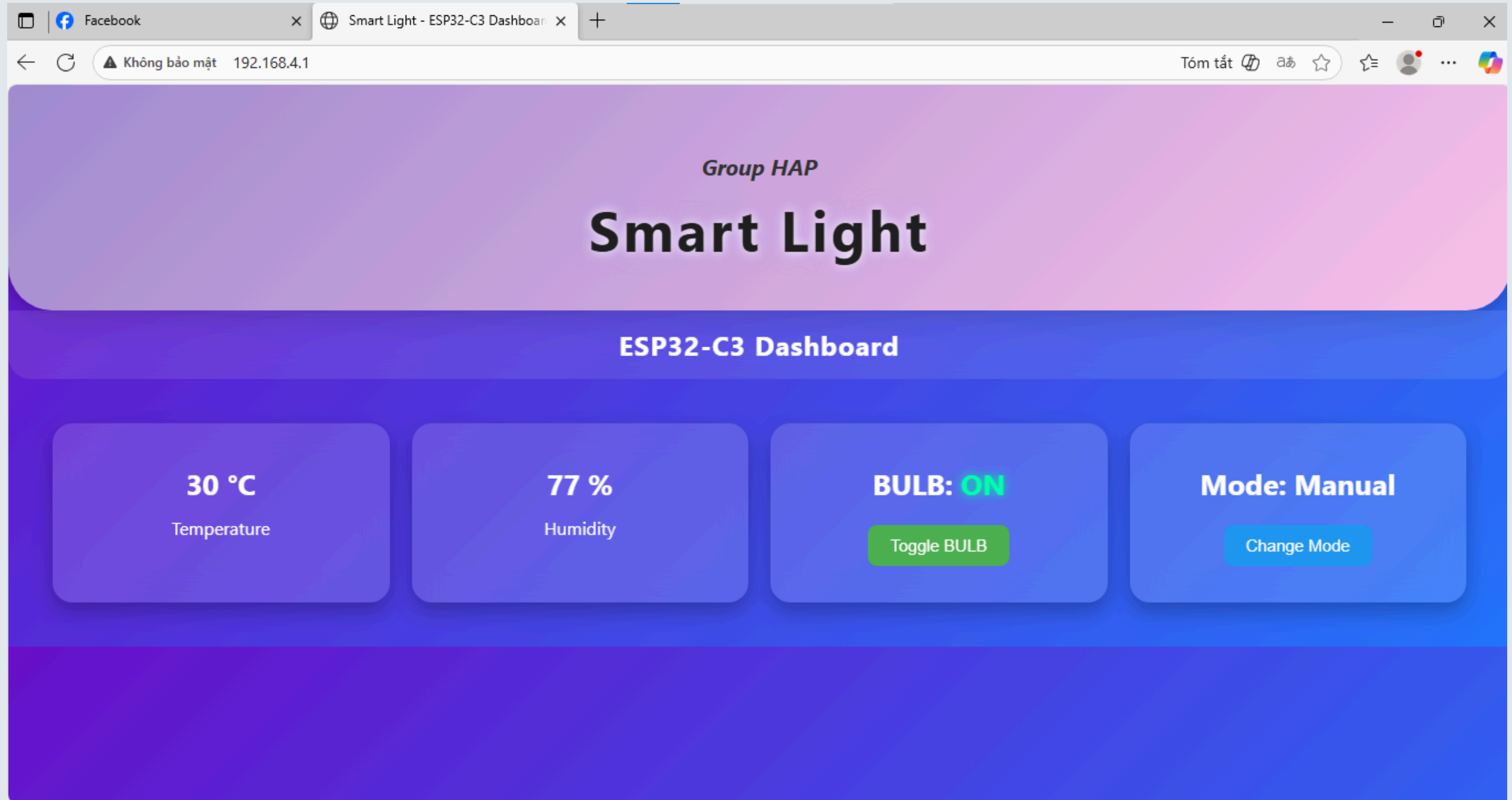
- Temperature
- Humidity
- Mode Light
- Status Light
- Light value: 0 - > 4095 (bright - > dark)



4. Evaluate the results

4.1 Result

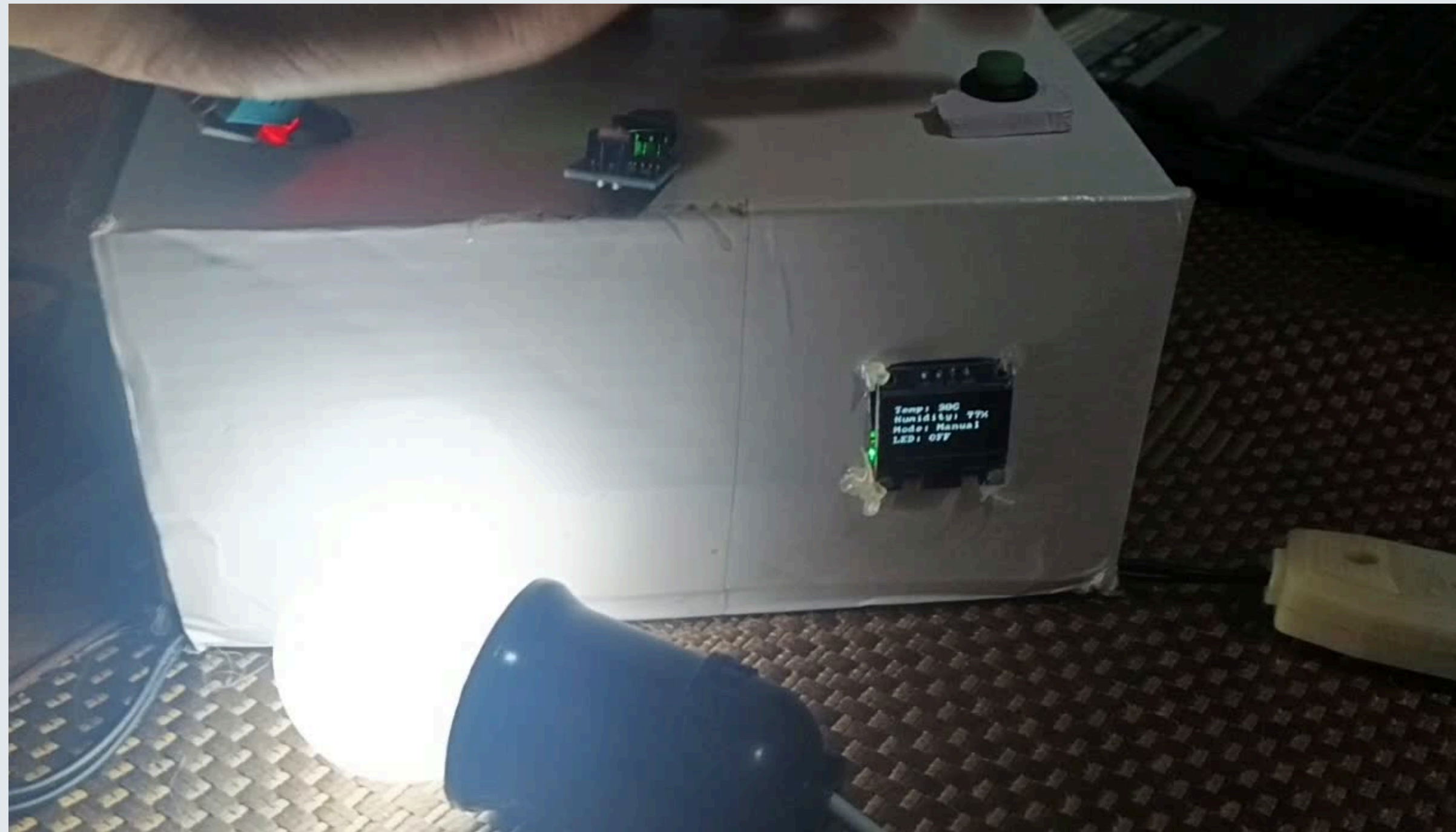
- Web server interface



4. Evaluate the results

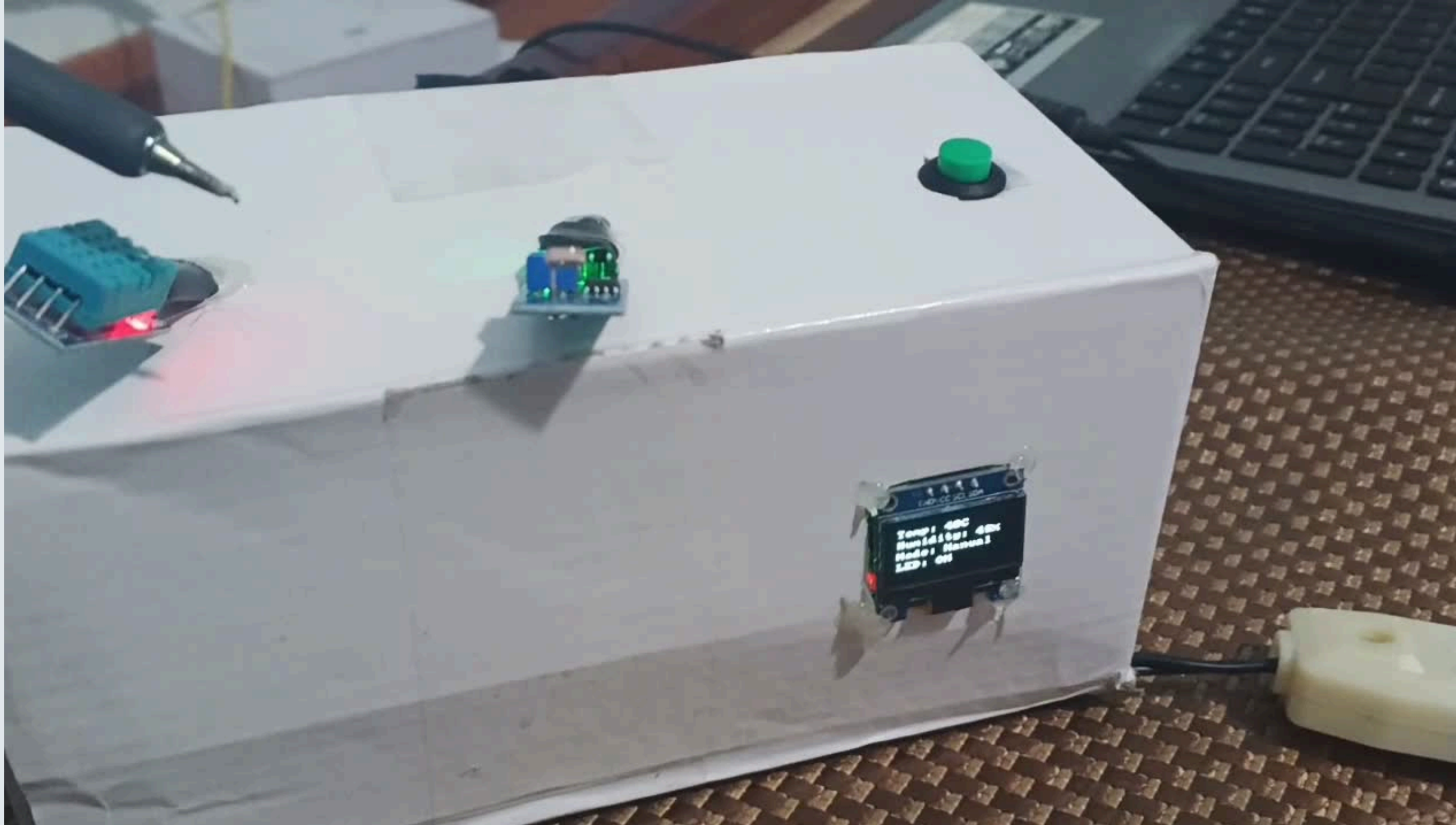
4.2 Demo

https://drive.google.com/drive/folders/1HIBwzW-jpdnXMSNGP_wI1lYhI1ggjtDY



4. Evaluate the results

4.2 Demo high temperature case ($>40^{\circ}\text{C}$)



THANK YOU FOR LISTENING